Year 11 Mathematics

Shape and Space 6

10.2 Triangles

Triangles

STARTER · 5 MIN

Date: _____ **Title:** Triangles

LEARNING OBJECTIVES

- Use angle facts in triangles and give reasons
- Solve angle problems in triangles using algebra
- Prove angle facts about triangles

STARTER *Answer in your book. No calculators.*

1. Angles on a straight line add up to?
2. Angles around a point add up to?
3. Solve $3x = 120$
4. An isosceles triangle has base angles of 50°. Find the top angle.

Starter — Answers

STARTER

Tick or correct in green pen

1. **180°**
2. **360°**
3. $x = \mathbf{40}$
4. $180 - 100 = \mathbf{80°}$

Success Criteria

SUCCESS CRITERIA

By the end of this lesson:

BRONZE	I can use angle facts in triangles and give reasons	☐
SILVER	I can solve angle problems in triangles using algebra	☐
GOLD	I can prove angle facts about triangles	☐

Tick each box in the plenary when you can do it on your own.

Introduction — Angles in Triangles

INTRODUCTION · 3 MIN

Every step in an angle problem needs a reason.

Angle sum

Angles in a triangle add up to 180°.

a + b + c = 180°

Isosceles

Two equal sides, two equal base angles.

Marked with dashes

Exterior angle

Equals the two opposite interior angles added.

d = a + b

Reasons

Write the angle fact you used.

angles on a straight line = 180°

Examples 1 and 2 — Angle Facts in Triangles

EXAMPLES · 5 MIN

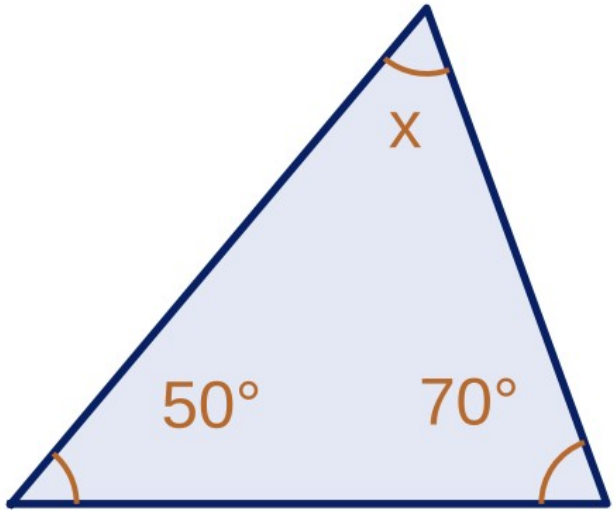
BRONZE

Watch and copy into your book

Hint triangle: 180° straight line: 180°

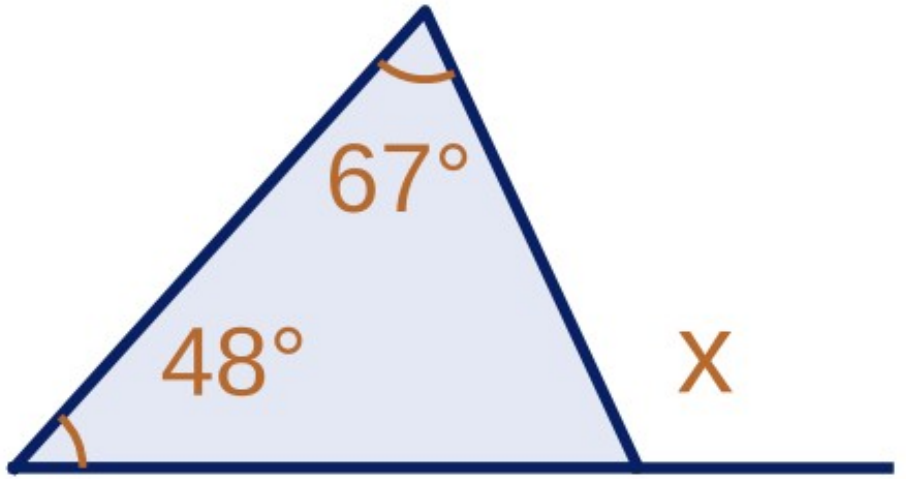
Example 1

Find x. Give a reason.



Example 2

Find the exterior angle x.



Examples 1 and 2 — Corrections

CORRECTIONS

BRONZE

Check your copy and correct in green pen

Example 1

$x = 180 - 50 - 70$

$x = 60°$

Angle sum of a triangle

Tip: always write the reason.

Example 2

$x = 48 + 67$

$x = 115°$

Exterior angle of a triangle

Tip: add the two opposite angles.

Task 1

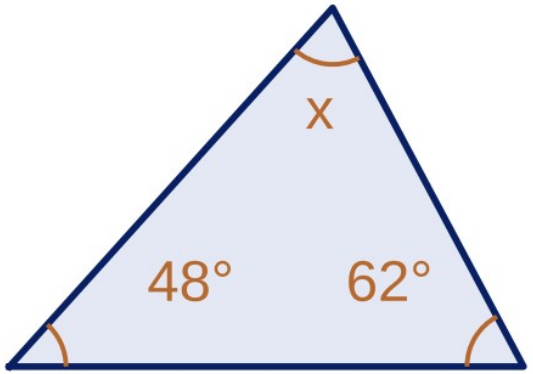
PRACTICE · 9 MIN

BRONZE

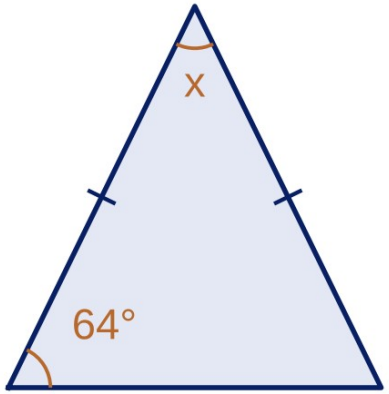
Find x. Give a reason.

Hint triangle: 180° straight line: 180°

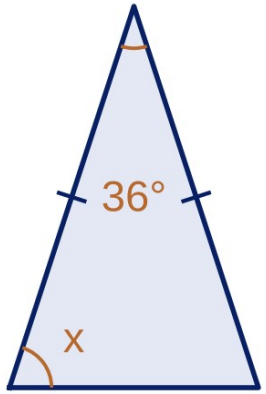
1



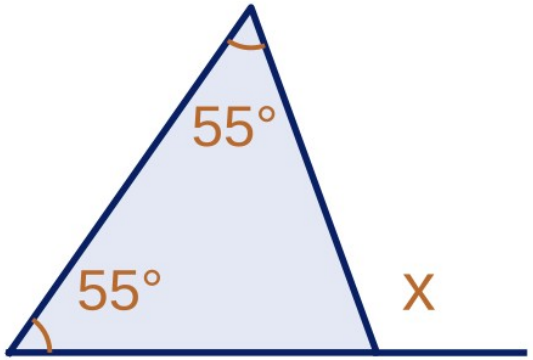
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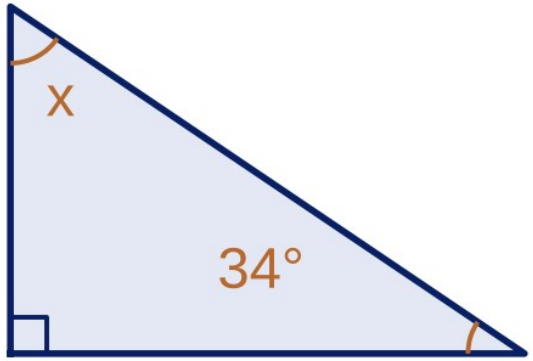
3



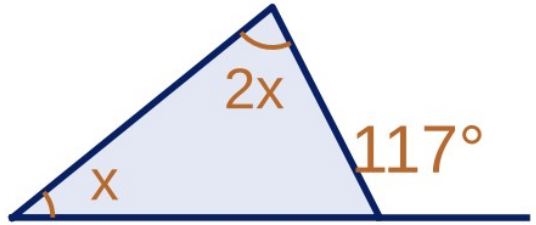
4



5



★



Task 1 — Answers

CORRECTIONS

BRONZE

Tick or correct in green pen

1. $180 - 48 - 62 = \mathbf{70°}$

2. $180 - 64 - 64 = \mathbf{52°}$

3. $(180 - 36) \div 2 = \mathbf{72°}$

4. $55 + 55 = \mathbf{125°}$

5. $180 - 90 - 34 = \mathbf{56°}$

★ $3x = 117, \quad x = \mathbf{39°}$

Examples 3 and 4 — Using Algebra

EXAMPLES · 5 MIN

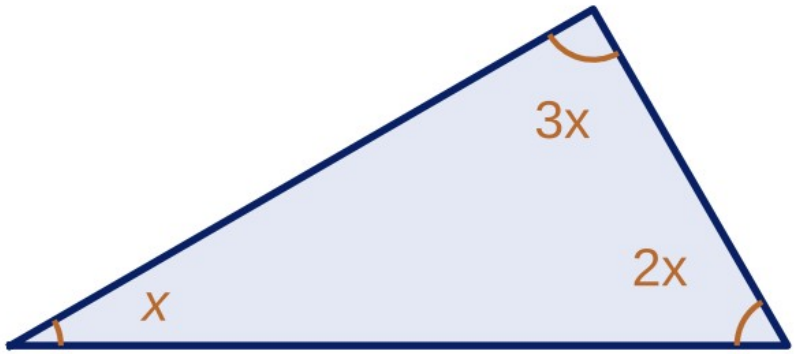
SILVER

Watch and copy into your book

Hint triangle: 180° isosceles: two equal angles

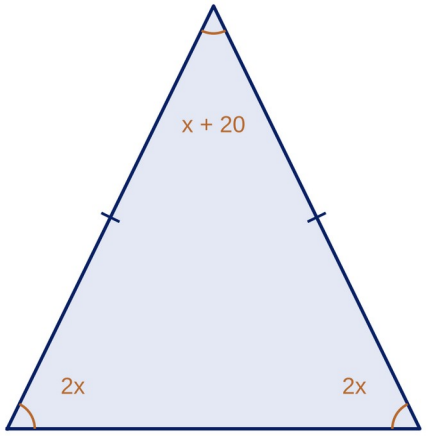
Example 3

Find x.



Example 4

Find x.



Examples 3 and 4 — Corrections

CORRECTIONS

SILVER

Check your copy and correct in green pen

Example 3

$$x + 2x + 3x = 180$$

$$6x = 180$$

$$\mathbf{x = 30°}$$

Tip: form an equation equal to 180.

Example 4

$$x + 20 + 2x + 2x = 180$$

$$5x = 160$$

$$\mathbf{x = 32°}$$

Tip: both base angles are 2x.

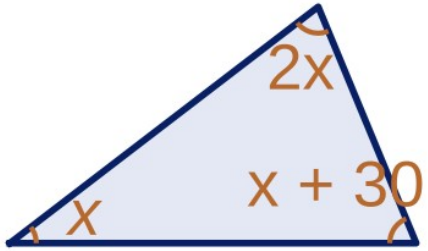
Task 2

PRACTICE · 9 MIN

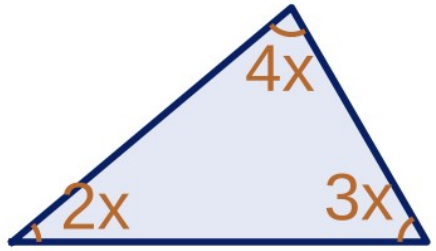
SILVER *Form an equation and solve it*

Hint triangle: 180° isosceles: two equal angles

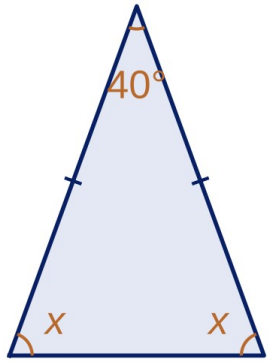
1 Find x.



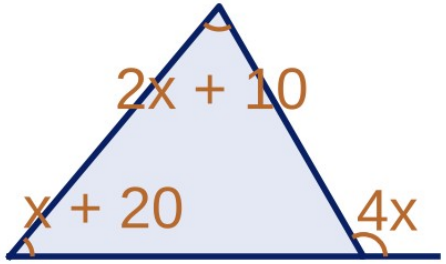
2 Find x.



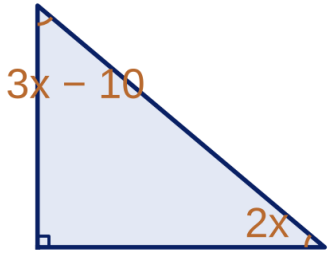
3 Find x.



4 Find x.



5 Find x.



Task 2 — Answers

CORRECTIONS

SILVER

Tick or correct in green pen

1. $4x + 30 = 180, \quad x = \mathbf{37.5°}$
2. $9x = 180, \quad x = \mathbf{20°}$
3. $2x + 40 = 180, \quad x = \mathbf{70°}$
4. $4x = 3x + 30, \quad x = \mathbf{30°}$
5. $5x - 10 = 90, \quad x = \mathbf{20°}$

Examples 5 and 6 — Reasons and Proofs

EXAMPLES · 5 MIN

GOLD

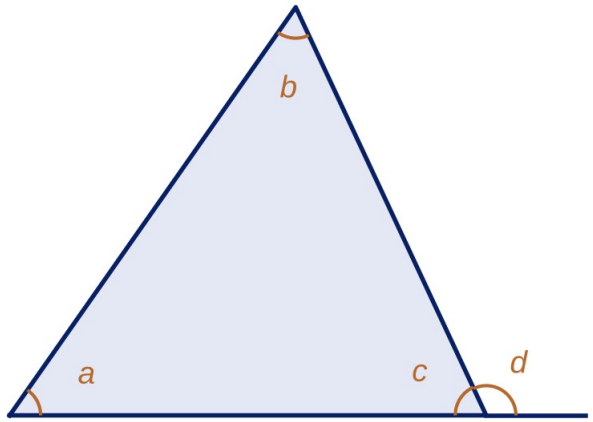
Exam-style: show all your working

Hint exterior angle = sum of the two opposite interior angles

Example 5

(3 marks)

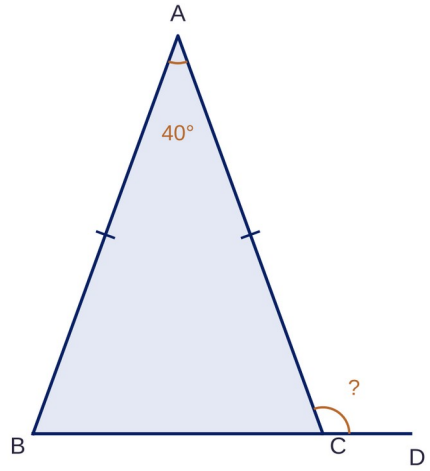
Prove that d = a + b.



Example 6

(3 marks)

AB = AC. Find angle ACD.



Examples 5 and 6 — Corrections

CORRECTIONS

GOLD

Check your copy and correct in green pen

Example 5

$a + b + c = 180$ (angle sum)

$c + d = 180$ (straight line)

So $d = a + b$

Tip: use two angle facts.

Example 6

$ACB = (180 - 40) \div 2$

$= 70°$ (isosceles)

$ACD = 180 - 70$ (straight line)

$ACD = 110°$

Tip: find the base angle first.

Task 3

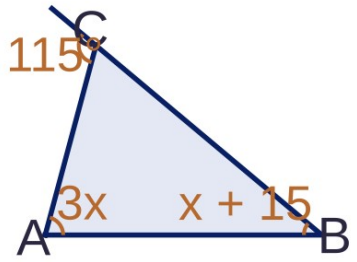
PRACTICE · 9 MIN

GOLD

Exam-style: show all your working

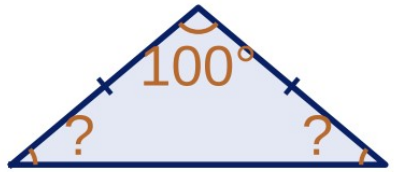
Hint exterior angle = sum of the two opposite interior angles

1 Find x.



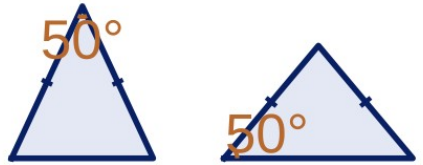
(3 marks)

2 Find the other two angles.



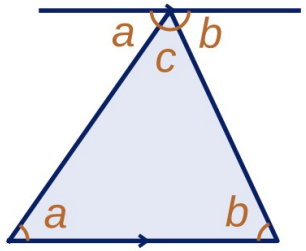
(2 marks)

3 Find the angles in each triangle.



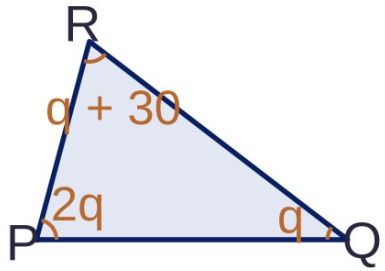
(3 marks)

4 Prove a + b + c = 180°.



(3 marks)

5 Find each angle.



(3 marks)

Task 3 — Answers

CORRECTIONS

GOLD

Tick or correct in green pen

1. $4x + 15 = 115, \quad x = \mathbf{25°}$

2. $(180 - 100) \div 2 = \mathbf{40°}$

3. **50°, 50°, 80° or 50°, 65°, 65°**

4. Two alternate angles, angles on a straight line = **180°**

5. $4Q + 30 = 180, \quad Q = 37.5°, \mathbf{37.5°}, \mathbf{75°}, \mathbf{67.5°}$

Plenary — Exit Ticket

PLENARY · 5 MIN

Answer in your book. Try all three.

BRONZE Two angles are 72° and 45°. Find the third.

SILVER The angles are x, 2x and 2x. Find x.

GOLD An exterior angle is 130°. One opposite interior angle is 75°. Find the other.

Reflect: Look back at the learning objectives. Which ones can you do now?

Plenary — Answers

PLENARY

Tick or correct in green pen

BRONZE $180 - 117 = \mathbf{63°}$

SILVER $5x = 180, \quad x = \mathbf{36°}$

GOLD $130 - 75 = \mathbf{55°}$

Exam Practice — Angle Facts in Triangles

EXAM PRACTICE · 10 MIN

BRONZE

IGCSE exam-style: show all your working

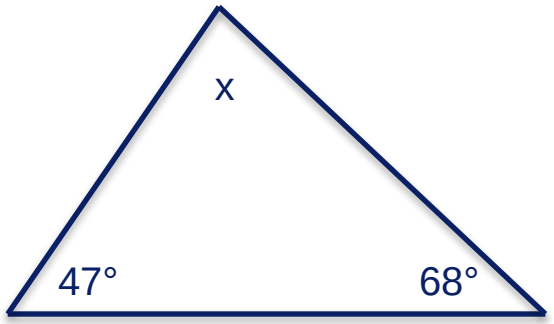
Hint triangle: 180° straight line: 180°

Question 1

(2 marks)

Find x. Give a reason.

Diagram NOT accurately drawn



x = ...

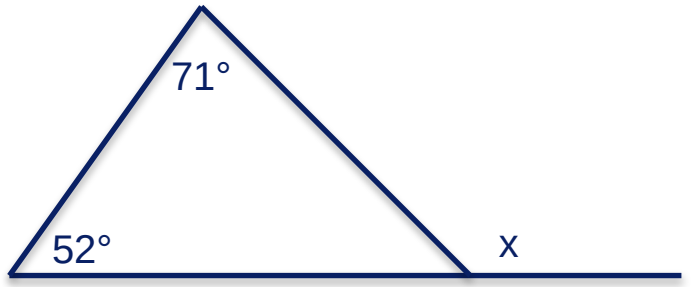
Question 2

(2 marks)

Find the exterior angle x.

Give a reason.

Diagram NOT accurately drawn



x = ...

Exam Practice — Using Algebra

EXAM PRACTICE · 10 MIN

SILVER

IGCSE exam-style: show all your working

Hint triangle: 180° isosceles: two equal angles

Question 3

(3 marks)

The angles of a triangle are x°, (2x + 10)° and (3x − 40)°.
Find x.

x = ...

Question 4

(3 marks)

An isosceles triangle has base angles of x° and a top angle of (2x + 20)°.
Find the size of each angle.

.....

Exam Practice — Reasons and Proofs

EXAM PRACTICE · 10 MIN

GOLD

IGCSE exam-style: show all your working

Hint exterior angle = sum of the two opposite interior angles

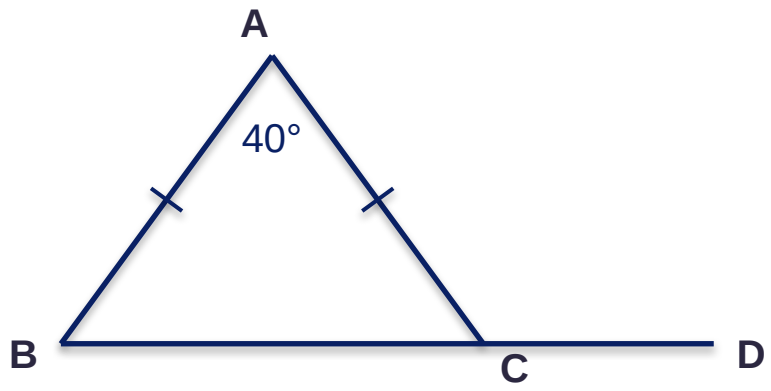
Question 5

(3 marks)

AB = AC. BCD is a straight line.

Find angle ACD. Give reasons.

Diagram NOT accurately drawn



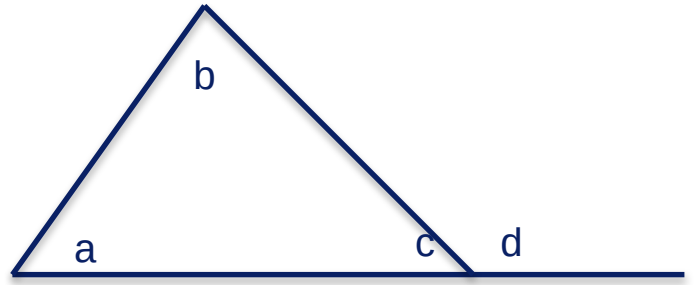
angle ACD = ...

Question 6

(3 marks)

Prove that d = a + b.

Diagram NOT accurately drawn



Exam Practice — Mark Scheme

CORRECTIONS

BRONZE

Mark your work: M = method mark, A = accuracy mark, B = independent mark

Question 1

x = 180 − 47 − 68 = 65° B1

angles in a triangle = 180° B1

Tip: always give the angle fact.

Question 2

x = 52 + 71 = 123° B1

exterior angle = sum of the B1

two opposite interior angles

Tip: or 180 − 57 = 123° using a straight line.

Exam Practice — Mark Scheme

CORRECTIONS

SILVER

Mark your work: M = method mark, A = accuracy mark, B = independent mark

Question 3

$x + 2x + 10 + 3x - 40 = 180$ — M1

$6x - 30 = 180$ — M1

$x = 35$ — A1

Tip: collect like terms first.

Question 4

$x + x + 2x + 20 = 180$ — M1

$4x = 160,\ x = 40$ — M1

40°, 40° and 100° — A1

Tip: an isosceles triangle has two base angles.

Exam Practice — Mark Scheme

CORRECTIONS

GOLD

Mark your work: M = method mark, A = accuracy mark, B = independent mark

Question 5

ABC = ACB = 140 ÷ 2 = 70° — M1

isosceles: base angles equal — B1

ACD = 180 − 70 = 110° — A1

(angles on a straight line)

Tip: the ticks show AB = AC.

Question 6

c + d = 180 (straight line) — B1

a + b + c = 180 (triangle) — B1

so d = a + b — B1

Tip: both are equal to 180 − c.